THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 52	$(10 \times 40 + 15 \times 60) \div 25$
2	**(a)** 42	Sum changes by 6
3	**(c)** 42.5	Arrange in ascending order first
4	**(c)** 40	Required total 180 minus 140
5	**(c)** 20	$k(n+1)/2 = 4 \times 10/2$
6	**(a)** 50%	Rise $= 10$; $(10/20)imes100$
7	**(b)** 41	$(20 \times 35 + 30 \times 45) \div 50$
8	**(b)** 36	Sum changes by 12
9	**(a)** 40.5	Arrange in ascending order first
10	**(d)** 0	Required total 150 minus 150

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total marks $= 15 \times 60 + 10 \times 70 = 1600$ (1); mean $= 64$ (1).	1+1
12	New mean $= 27 + 6 = 33$ (1+1).	1+1
13	Numbers: 3, 6, ..., 30 (1); mean $= k(n+1)/2 = 16.5$ (1).	1+1
14	Total marks $= 20 \times 45 + 30 \times 55 = 2550$ (1); mean $= 51$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total weight $= 10 \times 42 + 15 \times 48 = 1140$ kg (2); mean $= 45.6$ kg (1).	2+1
16	Sum changes by 10 (1); new sum $= 140$ (1); new mean $= 28$, change $= 2$ (1).	1+1+1
17	Median stays $= 49$ since the middle values are unchanged (1); mean increases by $152/7$ (1) to 72.71 (1).	1+1+1
	OR: Total weight $= 20 \times 36 + 30 \times 41 = 1950$ kg (2); mean $= 39$ kg (1).	2+1
18	Sum changes by 10 (1); new sum $= 185$ (1); new mean $= 37$, change $= 2$ (1).	1+1+1
	OR: Median stays $= 43$ since the middle values are unchanged (1); mean increases by $154/7$ (1) to 66.86 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) changes: 4, 5, 3, -4, -5 (2); (ii) mean $= 24$ (2); (iii) largest change 5°C (1).	2+2+1
	OR: Total $= 20 \times 50 + 30 \times 60 = 2800$ (2); mean $= 56$ (2); change $= 6$ (1).	2+2+1
20	(i) changes: 5, 6, 4, -5, -6 (2); (ii) mean $= 27.5$ (2); (iii) largest change 6°C (1).	2+2+1
	OR: Total $= 15 \times 64 + 10 \times 74 = 1700$ (2); mean $= 68$ (2); change $= 4$ (1).	2+2+1

Mistakes to avoid:

- Finding the median without arranging the data in order first.
- For an even number of observations, taking one of the middle values instead of the **mean of the two middle values**.
- Confusing mode with median or mean: mode is the most frequent value, not the middle value.
- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
- Computing range as the difference between the first and last values as listed, not the maximum and minimum.
- Reading a line graph at the wrong point or ignoring the scale of the axes.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 30	$k(n+1)/2 = 5 \times 12/2$
2	**(d)** 25%	Rise $= 10$; $(10/40)times100$
3	**(a)** 60	$(12 \times 50 + 8 \times 75) \div 20$
4	**(b)** 39	Sum changes by 18
5	**(a)** 73	Arrange in ascending order first
6	**(b)** 68	Required total 198 minus 130
7	**(c)** 24	$k(n+1)/2 = 3 \times 16/2$
8	**(a)** 20%	Rise $= 10$; $(10/50)times100$
9	**(d)** 30	$(25 \times 24 + 15 \times 40) \div 40$
10	**(a)** 44	Sum changes by 24

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New mean $= 23 + 9 = 32$ (1+1).	1+1
12	Numbers: 5, 10, ..., 60 (1); mean $= k(n+1)/2 = 32.5$ (1).	1+1
13	Total marks $= 12 \times 80 + 18 \times 70 = 2220$ (1); mean $= 74$ (1).	1+1
14	New mean $= 17 + 12 = 29$ (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total weight $= 8 \times 50 + 12 \times 45 = 940$ kg (2); mean $= 47$ kg (1).	2+1
16	Sum changes by 10 (1); new sum $= 105$ (1); new mean $= 21$, change $= 2$ (1).	1+1+1
17	Median stays $= 63$ since the middle values are unchanged (1); mean increases by $158/7$ (1) to 81.29 (1).	1+1+1
	OR: Total weight $= 16 \times 38 + 24 \times 43 = 1640$ kg (2); mean $= 41$ kg (1).	2+1
18	Sum changes by 10 (1); new sum $= 150$ (1); new mean $= 30$, change $= 2$ (1).	1+1+1
	OR: Median stays $= 43$ since the middle values are unchanged (1); mean increases by $154/7$ (1) to 63.43 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) changes: 2, 7, 5, -1, -5 (2); (ii) mean $= 22.67$ (2); (iii) largest change 7°C (1).	2+2+1
	OR: Total $= 24 \times 45 + 16 \times 60 = 2040$ (2); mean $= 51$ (2); change $= 6$ (1).	2+2+1
20	(i) changes: 4, 7, 3, -5, -6 (2); (ii) mean $= 28.83$ (2); (iii) largest change 7°C (1).	2+2+1
	OR: Total $= 30 \times 52 + 20 \times 62 = 2800$ (2); mean $= 56$ (2); change $= 4$ (1).	2+2+1

Mistakes to avoid:

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 65	Arrange in ascending order first
2	**(d)** 110	Required total 240 minus 130
3	**(d)** 42	$k(n+1)/2 = 6 \times 14/2$
4	**(b)** 25%	Rise $= 20$; $(20/80)times100$
5	**(d)** 24	$(30 \times 18 + 20 \times 33) \div 50$
6	**(d)** 32	Sum changes by 6
7	**(a)** 53.5	Arrange in ascending order first
8	**(b)** 37	Required total 132 minus 95
9	**(a)** 36	$k(n+1)/2 = 9 \times 8/2$
10	**(b)** 40%	Rise $= 10$; $(10/25)times100$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Numbers: 7, 14, ..., 63 (1); mean $= k(n+1)/2 = 35$ (1).	1+1
12	Total marks $= 25 \times 32 + 15 \times 48 = 1520$ (1); mean $= 38$ (1).	1+1
13	New mean $= 28 + 15 = 43$ (1+1).	1+1
14	Numbers: 2, 4, ..., 28 (1); mean $= k(n+1)/2 = 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total weight $= 25 \times 30 + 25 \times 34 = 1600$ kg (2); mean $= 32$ kg (1).	2+1
16	Sum changes by 10 (1); new sum $= 120$ (1); new mean $= 24$, change $= 2$ (1).	1+1+1
17	Median stays $= 41$ since the middle values are unchanged (1); mean increases by $136/7$ (1) to 61.29 (1).	1+1+1
	OR: Total weight $= 30 \times 28 + 20 \times 33 = 1500$ kg (2); mean $= 30$ kg (1).	2+1
18	Sum changes by 10 (1); new sum $= 145$ (1); new mean $= 29$, change $= 2$ (1).	1+1+1
	OR: Median stays $= 53$ since the middle values are unchanged (1); mean increases by $142/7$ (1) to 66.43 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) changes: 4, 7, 4, -5, -5 (2); (ii) mean $= 19.5$ (2); (iii) largest change 7°C (1).	2+2+1
	OR: Total $= 12 \times 70 + 18 \times 80 = 2280$ (2); mean $= 76$ (2); change $= 6$ (1).	2+2+1
20	(i) changes: 2, 9, 2, -3, -9 (2); (ii) mean $= 25.17$ (2); (iii) largest change 9°C (1).	2+2+1
	OR: Total $= 40 \times 35 + 10 \times 60 = 2000$ (2); mean $= 40$ (2); change $= 5$ (1).	2+2+1

Mistakes to avoid:

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